Postinjury Alcohol Use Is Associated With Prolonged Recovery After Concussion in NCAA Athletes

Roger C. Chang, MD, PhD,* Michael Singleton, PhD,† Sara P. D. Chrisman, MD,‡ Christopher C. Giza, MD,§ Ami Z. Cuneo, MD,* Natalia Murinova, MD,* Steven P. Broglio, PhD,¶ Michael McCrea, PhD,‖ Thomas W. McAllister, MD,** Tara L. Sharma, DO, MS,* and For the CARE Consortium Investigators

Abstract

Objective: To determine whether alcohol use leads to prolonged clinical recovery or increased severity of concussion symptoms in National Collegiate Athletic Association (NCAA) athletes. **Design:** Prospective observational study. **Setting:** Clinical institutions. **Participants:** Athletes from the NCAA Concussion Assessment Research and Education consortium who sustained a concussion from 2014 to 2021. **Interventions:** Athletes were divided into 2 groups, those reporting alcohol use postinjury and those reporting no alcohol use postinjury. **Main Outcome Measures:** Symptom recovery was evaluated as time (in days) from injury to clearance to return to unrestricted play (days until URTP). Severity of concussion symptoms was assessed using the Standardized Sport Concussion Assessment Tool (SCAT3) symptom severity, headache severity, difficulty concentrating, and difficulty remembering scores. These scores were taken a median of 6.6 [interquartile range (IQR) = 4.0-10] and 6 (IQR = 4.0-9.0) days after injury for those who did and did not consume alcohol postinjury respectively and compared with baseline SCAT3 scores. **Results:** Four hundred eighty four athletes from the data set had complete data for exposure and outcome. The adjusted mean number of days until URTP for athletes reporting alcohol use postinjury [23.3; 95% confidence interval (CI), 20.0-27.2; days] was incidence rate ratio (IRR) 1.32 (95% CI, 1.12-1.55; $P < 0.001$) times higher than for athletes who reported no alcohol use postinjury [17.7 (95% CI, 16.1-19.3) days]. Postinjury alcohol was not associated with severity of concussion symptoms (P's < 0.05). **Conclusion:** Self-reported postinjury alcohol use is associated with prolonged recovery but not severity of concussion symptoms in collegiate athletes. This may inform future clinical recommendations regarding alcohol consumption after concussion.

Key Words: concussion, alcohol use, return to play, traumatic brain injury, headache, depression, anxiety, collegiate athletes, Standardized Concussion Assessment Tool, Brief Symptom Inventory

(*Clin J Sport Med* 2023;00:1–8)

INTRODUCTION

Traumatic brain injury (TBI) can present with a range of clinical signs and symptoms including cognitive impairments, emotional lability, sleep disturbances, behavioral changes, somatic, and physical symptoms. Concussion, or mild TBI, accounts for 80% to 90% cases of traumatic brain injury.[1] Mild TBI rates across multiple sports have been increasing, leading to more attention to the impact of sports-related concussions.[2–4] Clinical symptoms in most people with mild TBI resolve within 2 to 4 weeks after injury; however, some studies show as many as 30% of individuals with mild TBI suffer from prolonged symptoms.[3–6] Therefore, understanding factors that can exacerbate concussion-related symptoms and lead to prolonged recovery is essential.

Although alcohol is known to increase the risk of TBI, the association between alcohol consumption and recovery after TBI has not been clearly elucidated. Studies with animal models suggest that acute low dose alcohol consumption before injury may offer neuroprotection through inhibiting NMDA receptors, reducing glucose metabolism uncoupling,

Submitted for publication November 18, 2022; accepted March 29, 2023.

*From the * University of Washington Neurology, Seattle, Washington † University of Washington, Institute of Translational Health Sciences, Seattle, Washington ‡ Seattle Children's Hospital, Seattle, Washington § University of California Los Angeles Steve Tisch BrainSPORT Program, Departments of Pediatrics and Neurosurgery, Los Angeles, California ¶ University of Michigan Concussion Center, Ann Arbor, Michigan ‖ Medical College of Wisconsin, Neurotrauma Research Center, Milwaukee, Wisconsin ** Indiana University School of Medicine Department of Psychiatry, Indianapolis, Indiana*

Data collection and sharing for this project was conducted with support from the Grand Alliance CARE Consortium, funded, in part by the NCAA and the Department of Defense. Moreover, the US Army Medical Research Acquisition Activity (820 Chandler St, Fort Detrick, MD 21702-5014) is the awarding and administering acquisition office. This work was supported by the Office of the Assistant Secretary of Defense for Health Affairs through the Psychological Health and Traumatic Brain Injury Program under award W81XWH-14- 2- 0151.

The authors report no conflicts of interest.

Corresponding Author: Tara L Sharma, 11011 Meridian Ave N Suite 201, Seattle, WA 98133, Email Address: tlsharma@uw.edu.

Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.cjsportmed.com).

Copyright © 2023 Wolters Kluwer Health, Inc. All rights reserved.

http://dx.doi.org/10.1097/JSM.0000000000001165

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

and restricting cell death.[1,7–13] In contrast, daily chronic alcohol consumption and/or acute alcohol intoxication may augment the excitotoxic effects of TBI through upregulation of NMDA receptors and reducing cerebral blow flow resulting in poorer outcomes.[7,12,–18] These data suggest that cognitive function and recovery after TBI may depend on the chronicity, timing, and amount of alcohol consumption.

Clinical studies evaluating the effect of alcohol use on recovery after TBI have yielded mixed results. Some have shown that alcohol use after mild or moderate TBI improves or has no effect on recovery outcomes.[19–21] Others suggest that alcohol consumption can worsen outcomes after TBI.[22,23] For example, alcohol consumption (primarily in patients with chronic alcohol use and/or acute intoxication) has been associated with longer episodes of posttraumatic amnesia, alterations in cerebral metabolism,[24–26] brain atrophy on magnetic resonance imaging, and difficulty with learning and memory after TBI with varying degrees of severity (mild, moderate, and severe).[27–34] Another study found that postinjury alcohol consumption was associated with worse executive functioning 6 to 9 months after severe TBI in subjects without a history of preinjury alcohol consumption.[29] No studies to date have investigated the effect of postinjury alcohol use on recovery and concussion symptom severity in a cohort of collegiate athletes.

In this study, we evaluated the interaction between postinjury alcohol use and concussion recovery. We also analyzed whether postinjury alcohol use affected severity of headache, concentration, and memory complaints, and total symptom burden after concussion. Headache severity was chosen as alcohol use has been shown to precipitate headache in some cases.[33] Difficulty concentrating and remembering were chosen as alcohol has been noted to result in poorer memory and concentration after TBI.[29,35] We hypothesized that postinjury alcohol consumption would prolong recovery, increase headache severity, lead to poorer concentration and memory, and increase total symptom burden after concussion.

METHODS

We used data from a prospective observational study conducted by the Concussion Assessment Research and Education (CARE) Consortium from August 2014 to November 2021 at 29 Clinical Study Core sites. The overall design and detailed methods of the CARE Consortium have been previously described.[30] Briefly, all concussions were diagnosed using the following definition: "a change in brain function following a force to the head, which may be accompanied by temporary loss of consciousness, but is identified in awake individuals with measures of neurologic and cognitive dysfunction."[31] Diagnosis of concussive events were completed by the medical staff at each site. All site personnel were trained on a standardized protocol for baseline testing and postinjury assessments before data collection.

Participants

The data set provided for this study included 4802 National Collegiate Athletic Association (NCAA) athletes and military service academy cadets with concussion. From this, we selected NCAA athletes with self-reported alcohol (ETOH) use status (Figure 1). For those with known alcohol use, only those who indicated drinking 1 or more alcoholic drinks per week during recovery were included. Questions about postinjury alcohol use were recorded at clearance to return to play and asked as follows, "has the athlete used alcohol since injury? If so, how many drinks per week?" Premorbid alcohol use was recorded at the baseline assessment and asked as follows, "Have you used alcohol in the past month? If yes, how many days per week over the past month did you drink?" Of note, athletes had the choice of directly answering these questions, leaving them blank, or skipping the question altogether. From the initial 4802 athletes with concussion in the sample, 484 had complete data for exposure and outcome (see Figure 1).

Ethical Considerations

Participation in this study was voluntary. All participants were informed of the purpose, benefits, and risks and written informed consent was obtained. The dataset was anonymized with each subject assigned a numerical ID. No personally identifiable data were available to the authors. All research activities at each participating site were approved by local institutional review boards and the Human Protection Research Office.

All consenting participants received a preseason/baseline assessment. This included demographics, medical history, baseline symptom scores (SCAT3), and psychological assessment (Brief Symptom Inventory-18 [BSI-18]); these assessments are described as follows:

- The Standardized Concussion Assessment Tool third edition (SCAT3) includes a 22-item self-report inventory designed to evaluate the total number and severity of common concussion symptoms on a 7-point Likert-type scale, (0 = none) to (6 = severe). Total symptom severity was evaluated using the total severity score from the SCAT3 (22 items × 6 levels give a range of 0-132). Headache severity, difficulty concentrating and remembering from the SCAT3 was determined by the 7-point Likert-type scale (range, 0-6).[30,32]
- The BSI-18 requires athletes to rate their level of distress over the past 7 days using a 5-point (ordinal) Likert-type scale, in which 0 is not at all and 4 is extremely often. This inventory provides scores on the following dimensions: somatization, depression, anxiety, and global severity index. Each of these clinical subscales comprises 6 items, so that the range of scores for each is 0 to 24.[30,34]

Participants were excluded if they had missing data of interest, alcohol use status was unknown, or they were non-NCAA athletes (military cadets not involved in NCAA level sports).

Concussion Recovery Outcome Measure

The CARE dataset has several outcome variables. Concussion recovery was evaluated as time from injury to when athlete was cleared to return to unrestricted play (days until URTP). This variable has been previously used in published studies to evaluate concussion recovery.[36,37]

Severity of Concussion Symptoms Outcome Measures

The SCAT3 outcome measures included total symptom severity and severity of 3 specific symptoms: headache, difficulty concentrating, and difficulty remembering. There were a series of SCAT3 scores recorded from the immediate postinjury timepoint (less than 6 hours after injury) until clearance to

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

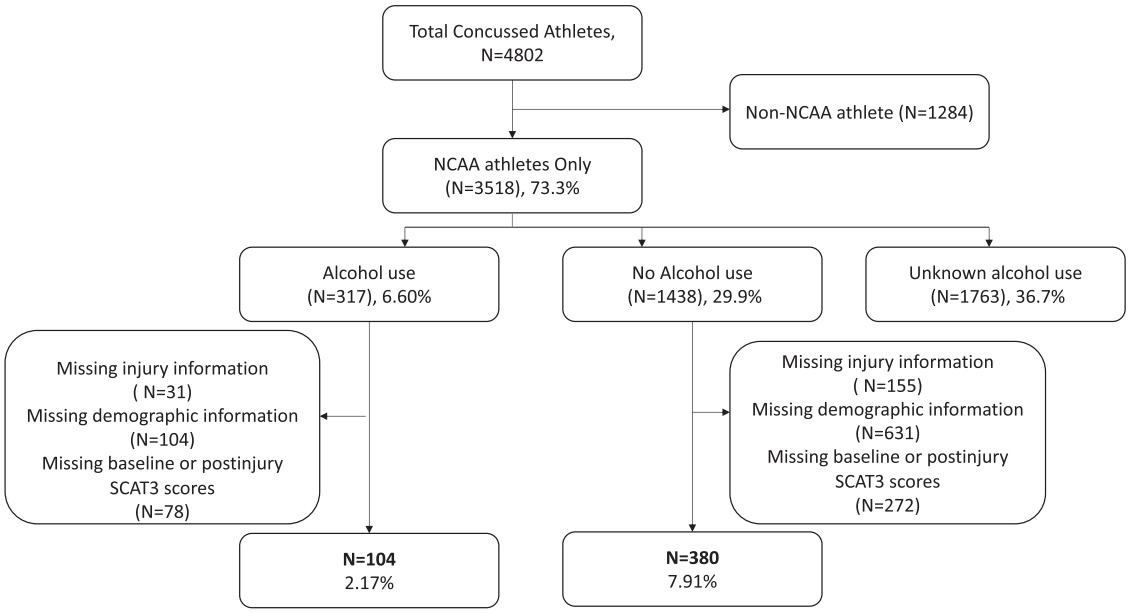

Figure 1. Total participants who met inclusion criteria. Number (N) and percent (%) of total concussed athletes.

return to unrestricted play (Figure 2). For our postinjury timepoint, we used the SCAT3 scores taken nearest to the asymptomatic score (SCAT3 postinjury scores) (Figure 2). This was an attempt to increase the likelihood that at least one alcoholic drink was consumed in those who reported postinjury alcohol use, while still being considered symptomatic postinjury. We also collected SCAT3 scores taken 24 to 48 hours after injury (SCAT3 24-48 hours postinjury scores) to evaluate whether SCAT3 outcome measures were similar between groups. To assess severity of concussion symptoms, we used percentage of athletes who returned to their baseline SCAT3 score (SCAT3 postinjury score was equal to their baseline score or better) at our chosen postinjury timepoint for each SCAT3 outcome measure described previously. Percentages were obtained by dividing the number of athletes who returned to their baseline SCAT3 score by the total number of athletes for each group (Table 1). This outcome measure was used to include baseline SCAT3 scores and to prevent extreme outliers from distorting the statistical analysis, which was apparent with change and raw scores.

Statistical Analysis

Univariate Analyses

Demographic variables including age,[38,39] sex,[40–42] sport type,[43] number of previous concussions,[43–45] preexisting migraine headaches,[46,47] postinjury substance use (marijuana, tobacco, and psychostimulant use), premorbid alcohol use as number of days per week alcohol was consumed in the past month (days per week drink), and baseline BSI depression, anxiety, and somatization scores[44,48] have been associated with recovery time in previous studies and thus were examined as potential confounders. Injury characteristics previously associated with recovery were also examined, including delayed reporting of symptoms, SCAT3 24 to 48 hours postinjury symptom severity scores, loss of consciousness (LOC), and posttraumatic amnesia.[36,37,49,50] For one of our primary outcome measures (severity of concussion symptoms), number of days after injury to SCAT3 postinjury scores (days to SCAT3 postinjury scores) were recorded, because this can influence SCAT3 postinjury outcome measures. These baseline demographic and injury variables were compared between those who reported alcohol use and those who reported no alcohol use using χ^2 (for categorical data), Fisher exact (for categorical data with small sample sizes), and independent t-tests (for continuous data). For t-tests, degrees of freedom were corrected when the Levene test indicated heterogeneity of variance between groups. Selected demographic variables were also compared between included athletes and the original population to assess for selection bias. Non-normally distributed data were presented as median [Interquartile range (IQR)]. IQR was defined as the second to third quartiles of the data set.

Multivariable Analyses

A multivariable negative binomial regression model was used to estimate the relative difference in mean URTP between groups. Poisson regression modeling strategy was used to account for the skew in distribution of this outcome variable. For severity of concussion symptoms outcome measures, a logistic regression model was used to estimate the odds ratio for returning to baseline SCAT3 in both groups. Observed confounding variables were adjusted for in each of these models. The procedure for selection of confounding variables in the final model followed the method outlined by Bursac et al (2008).[51] See Supplemental Digital Content (see Table 1, http://links.lww.com/JSM/A381) for the outline and statistical code for this selection process. Briefly, variables were only included if they significantly affected the association between alcohol use and outcome variables (evidence of confounding and $P<0.1$) in the presence of other tested covariates in the model (Step 2 and 3 of Table 1, Supplemental Digital Content, http://links.lww.com/JSM/A381).

A Spearman rho test was used to determine whether the number of postinjury alcoholic drinks per week correlated with days until URTP and SCAT3 postinjury outcome measures.

Hypothesis tests were two-sided with alpha = 0.05 significance level. All analyses were conducted using SPSS

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

TABLE 1. Demographic and Injury Characteristics for Athletes Who did Not (ETOH-) and Did Use Alcohol (ETOH+) Postinjury

Variable	ETOH- (n = 380)	ETOH+ (n = 104)	*P*
Age (M ± SD)	19.2 ± 1.35	19.5 ± 1.47	0.065
Days to SCAT3 postinjury scores (median [IQR])	6 [4-9]	6.6 [4-10]	0.492
Sex			
Female	185 (48.7)	47 (45.2)	0.528
Male	195 (51.3)	57 (54.8)	
Sport contact level			
Contact	259 (68.2)	69 (66.3)	0.228
Limited contact	87 (22.9)	20 (19.2)	
Noncontact	34 (8.9)	15 (14.4)	
Number previous concussions (M + SD)	0.75 ± 0.90	0.63 ± 0.74	0.240
LOC	14 (3.7)	11 (10.6)	0.005
PTA	44 (11.6)	10 (9.6)	0.573
Days symptoms reported late (M ± SD)	0.54 ± 1.02	0.80 ± 1.58	0.065
Depression	21 (5.5)	5 (4.8)	0.773
History of migraine headache	40 (10.5)	8 (7.7)	0.392
Premorbid alcohol use as number of days per week drink (M ± SD)	0.61 ± 1.11	1.60 ± 1.95	<0.001
Marijuana use postinjury*	2 (0.5)	10 (9.6)	<0.001
Tobacco use postinjury*	9 (2.4)	10 (9.6)	<0.001
Prescription medications, psychostimulant*	12 (3.2)	3 (2.9)	0.887
BSI Depression baseline scores (M ± SD)	0.87 ± 1.97	0.60 ± 1.30	0.183
BSI Anxiety baseline scores (M ± SD)	0.96 ± 2.07	0.73 ± 1.51	0.285
BSI Somatic symptoms baseline scores (M ± SD)	0.96 ± 2.13	0.81 ± 1.83	0.513
SCAT3 baseline raw scores (M + SD)			
SCAT3 symptom severity	5.36 ± 9.30	4.57 ± 7.32	0.361
SCAT3 headache severity	0.34 ± 0.856	0.37 ± 0.935	0.839
SCAT3 difficulty concentrating	0.34 ± 0.835	0.31 ± 0.925	0.725
SCAT3 difficulty remembering	0.25 ± 0.748	0.20 ± 0.597	0.517
SCAT3 return to baseline (%)			
SCAT3 symptom severity	246 (64.7)	66 (63.5)	0.81
SCAT3 headache severity	265 (69.7)	77 (74.0)	0.39
SCAT3 difficulty concentrating	320 (84.4)	90 (86.5)	0.60
SCAT3 difficulty remembering	351 (92.4)	99 (95.2)	0.32
SCAT3 24-48 hour postinjury raw scores (median [IQR])			
SCAT3 symptom severity	22 [9-43]	20 [9-45]	0.946
SCAT3 headache severity	2 [1-4]	2 [1-4]	0.857
SCAT3 difficulty concentrating	1 [1-3]	2 [2-3]	0.259
SCAT3 difficulty remembering	0 [0-1]	0 [0-2]	0.830

** Fisher exact tests were used instead of χ^2 test, because of small-expected counts.*
Values are reported as No. (%) unless otherwise indicated.

statistical software version 28 (IBM) and R software version 4.0.2 (R Core Team, 2020).[52,53]

RESULTS

Demographic and Injury Characteristics

Included athletes and the original population were similar except for statistically significant differences in age, sex, number of previous concussions, and SCAT3 symptom severity baseline scores (see Table 2, Supplemental Digital Content, http://links.lww.com/JSM/A382). There were significant differences between groups (postinjury alcohol use vs. no postinjury alcohol use) in LOC, premorbid alcohol use, and postinjury marijuana and tobacco use (see Table 1 for demographic comparisons). For the analysis of severity of concussion symptoms outcome measures, median number of days to SCAT3 postinjury scores was 6.6 (IQR = 4.0-10) and 6 (IQR = 4.0-9.0) for those who did and did not consume

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

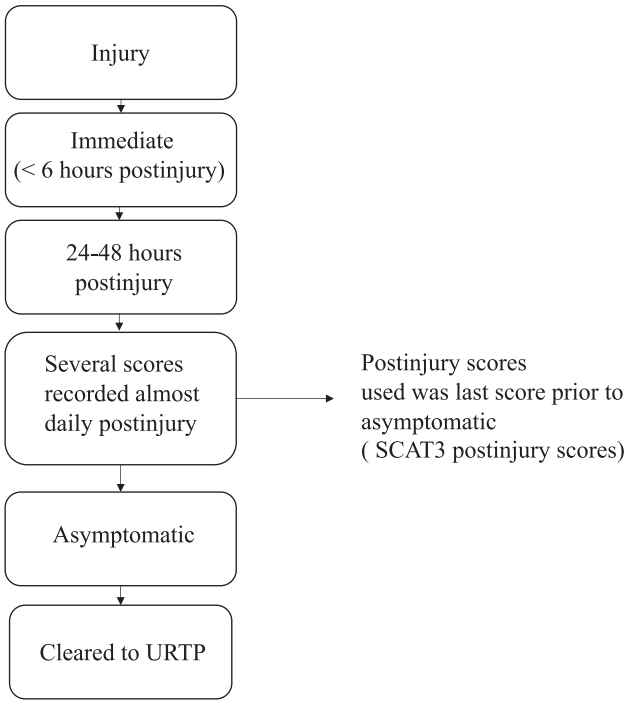

Figure 2. SCAT3 scores recorded for each athlete from injury until clearance to return to unrestricted play. SCAT3 postinjury scores used was the last score taken before athlete reporting being asymptomatic from concussion symptoms (about 1 day before asymptomatic score).

alcohol postinjury, respectively. This was also not significantly different between groups (Table 1).

Concussion Recovery Outcome Measure

Before adjustment, the mean number of days until URTP for athletes who reported alcohol use postinjury [22.0 (95% CI, 19.5-24.6) days] was incidence rate ratio (IRR) 1.33 (95% CI, 1.13-1.56) times higher than the mean number of days for athletes who reported no alcohol use postinjury [16.6 (95% CI, 15.2-17.9) days] (Figure 3). Age, delayed reporting of symptoms, and level of contact were determined to be confounders using the covariate selection procedure outlined in Supplemental Digital Content (see Table 1, http://links.lww.com/JSM/A381) and thus included in the final multivariable analysis. After adjusting for these variables, the mean number of days [23.3 (95% CI, 20.0-27.2) days] until URTP for athletes who reported alcohol use postinjury remained IRR 1.32 [95% CI (1.12-1.55); $P < 0.001$] times higher than for athletes who reported no alcohol use postinjury (17.7 [95% CI, 16.2-19.4] days). There was also a significant correlation between number of drinks per week and days until URTP [rho = 0.169; (95% CI, 0.078-0.256); $P < 0.001$] (Figure 4).

Severity of Concussion Symptoms Outcome Measures

When comparing athletes with alcohol exposure and those with no exposure, there was no significant difference in the proportion of athletes who returned to baseline SCAT3 symptom severity score during the study after adjusting for significant confounding variables [OR, 0.96 (95% CI, 0.87-1.07); Table 2]. Similarly, there was no significant difference between the 2 groups in return to baseline for SCAT3 difficulty remembering (OR 1.02; 95% CI, 0.96-1.08; Table 2), headache severity (OR, 1.00; 95% CI, 0.90-1.11; Table 2), or difficulty concentrating scores (OR, 0.99; 95% CI, 0.99-1.04; Table 2). There was also no significant correlation between number of drinks per week and SCAT3 postinjury symptom severity (rho = −0.071; 95% CI, −0.17-0.018; $P = 0.117$), SCAT3 headache severity (rho = −0.074; 95% CI, −0.164-0.018; $P = 0.105$), SCAT3 difficulty concentrating (rho = −0.039; 95% CI, −0.130-053; $P = 0.398$), and difficulty remembering (rho = −0.053; 95% CI, −0.144-0.039; $P = 0.246$) scores.

DISCUSSION

This study evaluated the association between postinjury alcohol use on concussion recovery and severity of concussion symptoms in NCAA athletes, finding that self-reported alcohol use after concussion was significantly associated with longer recovery. Consuming a greater number of drinks per week also correlated with slower recovery. These findings highlight the interaction between postinjury alcohol use and recovery after concussion.

Other studies have similarly shown that alcohol consumption can worsen clinical recovery after TBI. Outcomes have included increased agitation, extended hospital stays, and longer duration of post-traumatic amnesia.[23,54,55] For instance, several studies have shown increased pro-inflammatory chemokine expression along with astrocyte and microglial activation in alcohol intoxicated animals after mTBI.[56–58] These conditions can exacerbate the already elevated inflammatory state observed after a concussion and delay recovery.[12,56–59] In addition, athletes who reported postinjury alcohol use were significantly more likely to have also reported premorbid alcohol use, which could potentially increase inflammation, sensitization, and excitotoxic effects from concussion.[29,56–59] This premorbid alcohol consumption may also lead to poorer coping skills, predisposing them to developing prolonged symptoms after injury.

In addition to increased neuroinflammation, impaired cerebral blood flow autoregulation may explain worsened clinical recovery in alcohol users after concussion. For example, one study showed reduced cerebral blood flow, increased lactate concentrations, and reduced survival time in alcohol-treated swine after TBI.[7] Clinical studies have also

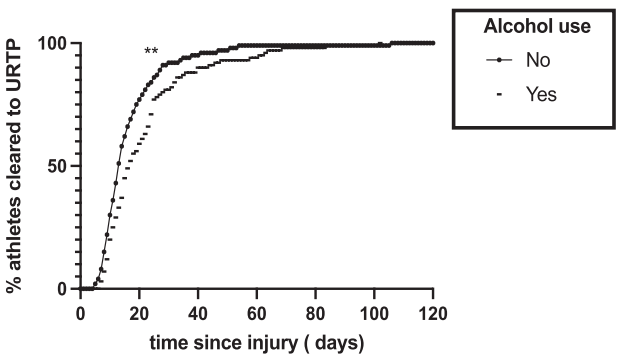

Figure 3. Recovery trajectories of the percentage (%) of athletes cleared to full unrestricted return to play (URTP) based on time since injury (days). $^{**}P < 0.001$.

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

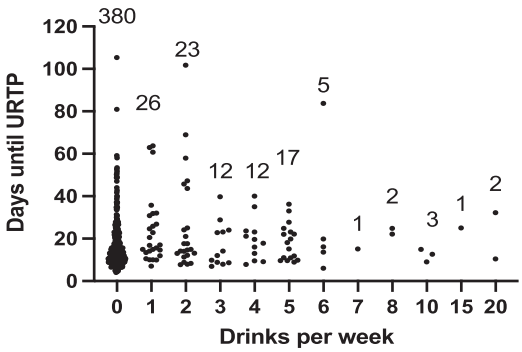

Figure 4. Scatter plot comparing days until URTP for each number of drinks per week. Numbers above the scatter indicate number of athletes for each number of drinks per week.

shown that alcohol consumption can cause fluctuations in cerebral blood flow in individuals without concussion.[60] Alcohol intake is postulated to increase the disrupted cerebral blood flow regulation observed after concussion and increase postconcussive symptoms.[61] Further studies analyzing the effects of alcohol use on cerebral blood flow after concussion using functional magnetic resonance imaging are needed.

In the secondary analysis, postinjury alcohol use did not influence total symptom severity or severity of individual symptoms (headache, difficulty concentrating, difficulty remembering). One explanation is that athletes may underplay symptoms in attempt to return to sport quickly. This leads to lower self-reported symptom scores. We also did not know exactly when athletes consumed alcohol after injury. Therefore, it is possible that some athletes who reported postinjury alcohol use had not yet consumed alcohol when their SCAT3 scores were obtained. This could have resulted in less accurate results. Interestingly, other studies have had similar results. For example, one study showed that preinjury, acute alcohol intoxication, and postinjury alcohol use did not influence symptom severity or neurocognitive outcomes a year after mild-to-moderate TBI.[19] Another showed that postinjury alcohol use did not affect postconcussion symptom scores 4 weeks after injury.[21] Similar to our study, these findings were limited by recall bias and self-reported symptom scores.

Limitations

There were some limitations to our study. One is that we cannot determine causality given the observational nature of this study. Another is that alcohol, tobacco, and marijuana use were self-reported. Furthermore, quantity of alcohol consumption was not standardized, which could have led athletes to have different interpretations of what constitutes one drink. Also, very few athletes reported drinking more than 5 alcoholic drinks per week (Figure 4). Therefore, it was difficult to accurately determine whether drinking greater than 5 drinks per week was associated with days until URTP or SCAT3 postinjury scores. Next, we were not able to compare immediate postinjury SCAT3 symptom scores (scores taken <6 hours after injury) between groups because of incomplete

TABLE 2. SCAT3 Outcome Variables Reported as Odds Ratio for Returning to Baseline SCAT3 Score Along With Covariates for Each Model

Variables	Odds Ratio Estimate	95% CI Range (Lower-Upper)	*P*
SCAT3 symptom severity			
Alcohol use postinjury (yes vs. no)	0.96	0.87-1.07	0.49
Zero score at baseline (yes vs. no)	0.74	0.68-0.81	<0.001
Number of days per week drink	1.02	0.99-1.05	0.20
Marijuana use postinjury (yes vs. no)	1.15	0.87-1.50	0.32
Previous concussion (yes vs. no)	1.05	1.00-1.10	0.04
BSI depression score	1.01	0.99-1.04	0.28
SCAT3 headache severity			
Alcohol use postinjury (yes vs. no)	1.00	0.90-1.11	0.98
Zero score at baseline (yes vs. no)	0.76	0.68-0.84	<0.001
Number of days per week drink	1.01	0.98-1.04	0.38
Tobacco use postinjury (yes vs. no)	1.21	0.99-1.49	0.07
Marijuana use postinjury (yes vs. no)	1.23	0.94-1.60	0.13
SCAT3 difficulty concentrating			
Alcohol use postinjury (yes vs. no)	0.99	0.99-1.04	0.90
Zero score at baseline (yes vs. no)	0.87	0.80-0.94	<0.001
Number of days per week drink	1.01	0.99-1.04	0.26
Loss of consciousness (yes vs. no)	1.08	0.94-1.25	0.28
Marijuana use postinjury (yes vs. no)	1.13	0.92-1.40	0.25
SCAT3 difficulty remembering			
Alcohol use postinjury (yes vs. no)	1.02	0.96-1.08	0.47
Zero score at baseline (yes vs. no)	0.94	0.88-1.01	0.07
Number of days per week drink	1.01	0.99-1.03	0.38

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

data. These missing data could be enlightening by showing whether individuals who experience greater physical symptoms immediately after concussion feel less inclined to drink alcohol or engage in social interactions. Conversely, those who experience more emotional symptoms after concussion may use alcohol to self-medicate their symptoms. Nevertheless, there were no significant differences between groups in the SCAT3 24 to 48 hours postinjury scores, and one would expect differences at 6 hours postinjury to also remain at 24 to 48 hours. Another limitation is that a large portion of athletes were excluded because of missing data. Despite this, selection bias within our sample was minimal (see Table 2, Supplemental Digital Content, http://links.lww.com/JSM/A382). Finally, information on whether episodic binge drinking versus mild and moderate drinking was associated with different concussion outcomes was not obtained.

CONCLUSIONS

This study demonstrated that postinjury alcohol use was associated with prolonged recovery, which suggests that avoiding alcohol consumption acutely after injury may shorten time to return to unrestricted play. We did not find any differences in symptom severity after concussion in athletes who consumed alcohol compared with those who did not. Additional studies could evaluate the effect of chronic alcohol consumption on concussion risk, along with long-term functional outcomes after concussion. Daily evaluations of symptom severity from injury until recovery for each group may also offer more insight into differences in symptom severity between groups. Overall, these findings will help clinicians educate athletes on alcohol use and its association with longer recovery times after concussion.

Appendix

The Concussion Assessment, Research, and Education (CARE) Consortium contributing Investigators include: April Marie (Reed) Hoy (Azusa Pacific University); Joseph B. Hazzard, Jr (Bloomsburg University); Louise A. Kelly (California Lutheran University); Justus D. Ortega (Humboldt State University); Nicholas Port (Indiana University); Margot Putukian (Princeton University); T. Dianne Langford, and Ryan Tierney (Temple University); Gerald McGinty, (US Air Force Academy); Patrick O'Donnell, (US Coast Guard Academy); Kenneth Cameron, Megan Houston, and Steven Svoboda, (US Military Academy); Joshua Goldman (University of California–Los Angeles); Holly J. Benjamin (University of Chicago); Thomas Buckley, and Thomas W. Kaminski (University of Delaware); James R. Clugston (University of Florida); Julianne D. Schmidt (University of Georgia); Luis A. Feigenbaum (University of Miami); James T. Eckner (University of Michigan); Jason P. Mihalik (University of North Carolina at Chapel Hill); Jessica Dysart Miles (University of North Georgia); Scott Anderson (University of Oklahoma); Christina L. Master (University of Pennsylvania); Anthony P. Kontos (University of Pittsburgh Medical Center); Jeffrey J. Bazarian (University of Rochester); M. Alison Brooks (University of Wisconsin); Stefan Duma, and Steven Rowson (Virginia Tech); Christopher M. Miles (Wake Forest University); Brian H. Dykhuizen (Wilmington College); and Laura Lintner (Winston-Salem State University).

ACKNOWLEDGMENTS

The authors thank Jody Harland, MS, CIP, LSSBB (Indiana University) for her assistance in data management. The authors also thank the research and medical staff at each of the CARE participation sites.

References

1. Baratz R, Rubovitch V, Frenk H, et al. The influence of alcohol on behavioral recovery after mTBI in mice. *J Neurotrauma*. 2010;27(3): 555–563.
2. Tsushima WT, Siu AM, Ahn HJ, et al. Incidence and risk of concussions in youth athletes: comparisons of age, sex, concussion history, sport, and football position. *Arch Clin Neuropsychol*. 2019;34:60–69.
3. Daneshvar DH, Nowinski CJ, McKee AC, et al. The epidemiology of sport-related concussion. *Clin Sports Med*. 2011;30:1–17.
4. Zuckerman SL, Kerr ZY, Yengo-Kahn A, et al. Epidemiology of sports-related concussion in NCAA athletes from 2009-2010 to 2013-2014. *Am J Sports Med*. 2015;43:2654–2662.
5. Levin HS, Mattis S, Ruff RM, et al. Neurobehavioral outcome following minor head injury: a three-center study. *J Neurosurgery*. 1987;66(2): 234–243.
6. Broglio SP, McAllister T, Katz BP, et al. The natural history of sport-related concussion in collegiate athletes: findings from the NCAA-DoD CARE consortium. *Sports Med*. 2022;52:403–415.
7. Zink BJ, Stern SA, Wang X, et al. Effects of ethanol in an experimental model of combined traumatic brain injury and hemorrhagic shock. *Acad Emerg Med*. 1998;5:9–17.
8. Kelly DF, Kozlowski DA, et al. Ethanol reduces metabolic uncoupling following experimental head injury. *J Neurotrauma*. 2000;17(4): 261–272.
9. Gluckman S, Macgregor R. Effect of acute alcohol intoxication on granulocyte mobilization and kinetics. *Blood*. 1978;52:551–559.
10. Biegon A, Fry PA, Paden CM, et al. Dynamic changes in N-methyl-d-aspartate receptors after closed head injury in mice: implications for treatment of neurological and cognitive deficits. *Proc Natl Acad Sci*. 2004; 101:5117–5122.
11. Young AB, Ott LG, Beard D, et al. The acute-phase response of the brain-injured patient. *J Neurosurg*. 1988;69:375–380.
12. Zink BJ, Sheinberg MA, Wang X, et al. Acute ethanol intoxication in a model of traumatic brain injury with hemorrhagic shock: effects on early physiological response. *J Neurosurg*. 1998;89:983–990.
13. Nagy J Alcohol related changes in regulation of NMDA receptor functions. *Curr Neuropharmacology*. 2008;6:39–54.
14. Berglund M, Hagstadius S, Risberg J, et al. Normalization of regional cerebral blood flow in alcoholics during the first 7 weeks of abstinence. *Acta Psychiatr Scand*. 1987;75:202–208.
15. Caspari D, Trabert W, Heinz G, et al. The pattern of regional cerebral blood flow during alcohol withdrawal - a single photon emission tomography study with ^{99m}Tc-HMPAO. *Acta Med Scand*. 1993;87:414–417.
16. Dally S, Luft A, Ponsin JC, et al. Abnormal pattern of cerebral blood flow distribution in young alcohol addicts. *Br J Addict*. 1988;83:105–109.
17. Dupont RM, Rourke SB, Grant I, et al. Single photon emission computed tomography with iodoam-phetamine-123 and neuropsychological studies in long-term abstinent alcoholics. *Psychiatr Res Neuroimaging*. 1996;67: 99–111.
18. Hata T, Meyer JS, Tanahashi N, et al. Three-dimensional mapping of local cerebral perfusion in alcoholic encephalopathy with and without Wernicke-Korsakoff syndrome. *J Cereb Blood Flow Metab*. 1987;7:35–44.
19. Silverberg ND, Panenka W, Iverson GL, et al. Alcohol consumption does not impede recovery from mild to moderate traumatic brain injury. *J Int Neuropsychol Soc*. 2016;22:816–827.
20. Scheenen ME, De Koning ME, Van Der Horn HJ, et al. Acute alcohol intoxication in patients with mild traumatic brain injury: characteristics, recovery, and outcome. *J NeuroTrauma*. 2016;33(4):339–345.
21. Lawrence DW, Foster E, Comper P, et al. Cannabis, alcohol and cigarette use during the acute post-concussion period. *Brain Inj*. 2020; 34:42–51.
22. O'Phelan K, McArthur DL, Chang CW, et al. The impact of substance abuse on mortality in patients with severe traumatic brain injury. *J Trauma*. 2008;65(3):674–677.
23. Kaplan CP, Corrigan JD. Effect of blood alcohol level on recovery from severe closed head injury. *Brain Inj*. 1992;6:337–349.

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.

24. Adams KM, Gilman S, Koeppe RA, et al. Neuropsychological deficits are correlated with frontal hypometabolism in positron emission tomography studies of older alcoholic patients. *Alcohol Clin Exp Res*. 1993;17: 205–210.
25. Gilman S, Adams KM, Johnson-Greene D, et al. Effects of disulfiram on positron emission tomography and neuropsychological studies in severe chronic alcoholism. *Alcohol Clin Exp Res*. 1996;20:1456–1461.
26. Volkow ND, Hitzemann R, Wolf AP, et al. Acute effects of ethanol on regional brain glucose metabolism and transport. *Psychiatr Res*. 1990;35: 39–48.
27. Netrakom P, Kruaski JS, Miller NS, et al. Structural and functional neuroimaging findings in substance-related disorders. *Psychiatr Clin North Am*. 1999;22:313–329.
28. Pfefferbaum A, Sullivan EV, Rosenbloom MJ. A controlled study of cortical gray matter and ventricular changes in alcoholic men over a 5-year interval. *Arch Gen Psychiatr*. 1998;55:905–912.
29. Ponsford J, Tweedly L, Taffe J. The relationship between alcohol and cognitive functioning following traumatic brain injury. *J Clin Exp Neuropsychol*. 2013;35:103–112.
30. Broglio SP, Mccrea M, Mcallister T, et al. A National study on the effects of concussion in collegiate athletes and US military service academy members: the NCAA–DoD concussion assessment, research and education (CARE) consortium structure and methods. *Sports Med*. 2017;47:1437–1451.
31. Carney N, Ghajar J, Jagoda A, et al. Concussion guidelines step 1: systematic review of prevalent indicators. *Neurosurgery*. 2014;75(suppl 1):S3–S15.
32. McCrory P, Meeuwisse WH, Aubry M, et al. Consensus statement on concussion in sport: the 4th International Conference on concussion in sport held in Zurich, November 2012. *Br J Sports Med*. 2013;47: 250–258.
33. Davis-Martin R, Polk AN, Smitherman TA. Alcohol use as a comorbidity and precipitant of primary headache: review and meta-analysis. *Curr Pain Headache Rep*. 2017;21:42.
34. Meachen SJ, Hanks RA, Millis SR, et al. The reliability and validity of the brief symptom inventory-18 in persons with traumatic brain injury. *Arch Phys Med Rehabil*. 2008;89:958–965.
35. Wilde EA, Bigler ED, Gandhi PV, et al. Alcohol abuse and traumatic brain injury: quantitative magnetic resonance imaging and neuropsychological outcome. *J Neurotrauma*. 2004;21:137–147.
36. Asken BM, Bauer RM, Guskiewicz KM, et al. Immediate removal from activity after sport-related concussion is associated with shorter clinical recovery and less severe symptoms in collegiate student-athletes. *Am J Sports Med*. 2018;46:1465–1474.
37. Asken BM, McCrea MA, Clugston JR, et al. Playing through it: delayed reporting and removal from athletic activity after concussion predicts prolonged recovery. *J Athletic Train*. 2016;51:329–335.
38. Covassin T, Elbin RJ, Harris W, et al. The role of age and sex in symptoms, neurocognitive performance, and postural stability in athletes after concussion. *Am J Sports Med*. 2012;40:1303–1312.
39. Zuckerman SL, Lee YM, Odom MJ, et al. Recovery from sports-related concussion: days to return to neurocognitive baseline in adolescents versus young adults. *Surg Neurol Int*. 2012;3:130.
40. Sicard V, Moore RD, Ellemberg D. Long term cognitive outcomes in male and female athletes following sport-related concussions. *Int J Psychophysiol*. 2018; 132:3–8.
41. Covassin T, Elbin R, Kontos A, et al. Investigating baseline neurocognitive performance between male and female athletes with a history of multiple concussion. *J Neurol Neurosurg Psychiatr*. 2010;81:597–601.
42. Merritt VC, Padgett CR, Jak AJ. A systematic review of sex differences in concussion outcome: what do we know? *Clin Neuropsychol*. 2019;33: 1016–1043.
43. Hume PA, Theadom A, Lewis GN, et al. A comparison of cognitive function in former rugby union players compared with former non-contact-sport players and the impact of concussion history. *Sports Med*. 2017;47:1209–1220.
44. Corwin DJ, Zonfrillo MR, Master CL, et al. Characteristics of prolonged concussion recovery in a pediatric subspecialty referral population. *J Pediatr*. 2014;165:1207–1215.
45. Abrahams S, Fie SM, Patricios J, et al. Risk factors for sports concussion: an evidence-based systematic review. *Br J Sports Med*. 2014;48:91–97.
46. Iverson GL, Silverberg ND, Mannix R, et al. Factors associated with concussion-like symptom reporting in high school athletes. *JAMA Pediatr*. 2015;169:1132–1140.
47. Hoffman JM, Lucas S, Dikmen S, et al. Natural history of headache after traumatic brain injury. *J Neurotrauma*. 2011;28:1719–1725.
48. Iverson GL, Gardner AJ, Terry DP, et al. Predictors of clinical recovery from concussion: a systematic review. *Br J Sports Med*. 2017;51:941–948.
49. Teel EF, Marshall SW, Shankar V, et al. Predicting recovery patterns after sport-related concussion. *J Athl Train*. 2017;52:288–298.
50. Kelly JP. Loss of consciousness: pathophysiology and implications in grading and safe return to play. *J Athl Train*. 2001;36:249–252.
51. Bursac Z, Gauss CH, Williams DK, et al. Purposeful selection of variables in logistic regression. *Source Code Biol Med*. 2008;3:17.
52. *IBM SPSS Statistics for Windows, Version 28.0*. Armonk, NY: IBM Corp; 2021.
53. R Core Team. *A Language and Environment for Statistical Computing*. Vienna, Austria: R Foundation for Statistical Computing, 2020.
54. Brooks N, Symington C, Beattie A, et al. Alcohol and other predictors of cognitive recovery after severe head injury. *Brain Inj*. 1989;3:235–246.
55. Jorge RE, Starkstein SE, Arndt S, et al. Alcohol misuse and mood disorders following traumatic brain injury. *Arch Gen Psychiatr*. 2005;62:742.
56. S Tate P, David MF, Charles HB, et al. Traumatic brain injury: influence of blood alcohol level on post-acute cognitive function. *Brain Inj*. 1999;13: 767–784.
57. Tulloh BR, Collopy BT. Positive correlation between blood alcohol level and ISS in road trauma. *Injury*. 1994;25:539–543.
58. Sabhesan S, Natarajan M. Post traumatic amnesia and alcohol dependence. *Indian J Psychiatr*. 1988;30:345–348.
59. Sparadeo FR, Gill D. Effects of prior alcohol use on head injury recovery. *J Head Trauma Rehabil*. 1989;4:75–81.
60. Gundersen H, van Wageningen H, Grüner R. Alcohol-induced changes in cerebral blood flow and cerebral blood volume in social drinkers. *Alcohol Alcohol*. 2013;48:160–165.
61. Churchill NW, Hutchison MG, Graham SJ, et al. Symptom correlates of cerebral blood flow following acute concussion. *NeuroImage Clin*. 2017; 16:234–239.
58. QinL, CrewsFT Chronic ethanol increases systemic TLR3 agonist-induced neuroinflammation and neurodegeneration *J Neuroinflammation* 2012;9: 603.
59. Ghirnikar RS, Lee YL, Eng LF. Inflammation in traumatic brain injury: role of cytokines and chemokines. *Neurochem Res*. 1998;23:329–340.
60. Gundersen H, van Wageningen H, Grüner R. Alcohol-induced changes in cerebral blood flow and cerebral blood volume in social drinkers. *Alcohol Alcohol*. 2013;48:160–165.
61. Churchill NW, Hutchison MG, Graham SJ, et al. Symptom correlates of cerebral blood flow following acute concussion. *NeuroImage Clin*. 2017; 16:234–239.

Copyright © 2023 Wolters Kluwer Health, Inc. Unauthorized reproduction of this article is prohibited.